

CUSTOMER INTELLIGENCE PLATFORM

Complete Documentation & Project Report

Customer Personality Analysis | K-Means Customer Segmentation | Streamlit

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Document	Day 5 — Documentation & Presentation

1. Executive Summary

- The Customer Intelligence Platform is an end-to-end machine-learning application for customer personality analysis and behavioral segmentation.
- The project uses a trained K-Means clustering model and a Streamlit interface to provide customer segmentation, prediction, search, visualization, and marketing recommendations.
- The application dynamically searches the project directory for Final Dataset.csv, kmeans_model.pkl, and day3_results.pkl.

2. Project Objectives

- Prepare and analyze customer data for machine-learning use.
- Identify customer groups using unsupervised K-Means clustering.
- Save and reuse the trained clustering model for inference.
- Build an interactive Streamlit customer intelligence application.
- Provide customer search, segment analysis, prediction, visualization, and marketing insights.
- Perform quality assurance and prepare final documentation.

3. Technology Stack

- Python
- Pandas and NumPy
- Scikit-Learn
- Joblib
- Streamlit
- Pathlib and os for file discovery

4. Project Statistics

Metric	Value
Total customers	2,038
Total features shown by the application	42
Customer segments	2

Model	K-Means Unsupervised
Saved model	kmeans_model.pkl

5. Application Modules

- **Home** — platform overview, dataset statistics, system status, and detected files.
- **Dashboard** — customer counts, segment distribution, dataset preview, and numeric statistics.
- **Customer Segments** — selected-segment customer count, average income, average spending, and records.
- **Customer Search** — case-insensitive search across the final dataset.
- **Customer Prediction** — uses the saved K-Means model to predict an incoming customer's segment.
- **Visualizations** — selectable numeric-feature visualization and descriptive statistics.
- **Marketing Strategy** — loads saved Day 3 business recommendations.
- **Download** — downloads project outputs and displays detected files.

6. Model Inference & Feature Compatibility

- The application loads kmeans_model.pkl with Joblib.
- It checks feature_names_in_ when available and constructs the prediction row in the model's feature order.
- It also checks n_features_in_ so an incorrect feature count is not silently passed to the model.
- When a manually entered feature is missing, the application can complete it from dataset statistics.
- If exact preprocessing is unavailable, the application reports that the original training/preprocessing pipeline may be required.

7. Error Handling & Important Fixes

- A StreamlitValueBelowMinError occurred because an input used min_value=0 while a valid dataset default was negative.
- The numeric input helper was changed to use the real finite dataset minimum and maximum and clamp the default median inside that range.
- The application also handles missing files and prediction exceptions without terminating the complete interface.
- A notebook QA issue involving globals().items() was avoided by using safer dataframe-variable checks rather than modifying a dictionary during iteration.

8. Day 4 — Quality Assurance

- QA testing produced 6 total tests.
- Passed tests: 6.
- Failed tests: 0.
- QA artifact saved at D:\AI-Lab-99\Week 5\Day 4\qa_results.pkl.
- This result indicates that all six checks in the submitted QA result set passed.

9. Business Value

- Customer segmentation helps identify behavioral groups.
- Customer profiles can support targeted communication.
- Marketing recommendations can be aligned with segment characteristics.
- The prediction page demonstrates reuse of a trained clustering model for an incoming customer.
- The dashboard provides a single interface for analysis and decision support.

10. Limitations

- K-Means cluster IDs are labels, not inherently meaningful customer names.
- Prediction quality depends on matching the preprocessing and feature engineering used during training.
- The current application searches a local project directory and is configured for the development environment.
- Marketing recommendations are loaded from saved Day 3 results rather than generated by a separate real-time recommendation model.

11. Future Improvements

- Save the complete preprocessing and K-Means pipeline as one production artifact.
- Add model versioning and training metadata.
- Add automated smoke tests for application pages.
- Use environment-based configuration instead of a fixed local path.
- Add stronger authentication, monitoring, and deployment configuration.
- Add richer interactive visualizations and downloadable segment reports.

12. Final Submission Checklist

- Final Dataset.csv
- kmeans_model.pkl
- day3_results.pkl
- qa_results.pkl
- Streamlit application
- Day 4 QA results
- Complete documentation
- Presentation

13. Conclusion

- The project integrates customer data analysis, unsupervised machine learning, business insights, quality assurance, and an interactive Streamlit application.
- Day 4 QA achieved 6/6 passed tests with 0 failures.
- Day 5 packages the completed technical work into a formal report and presentation suitable for submission.

14. Suggested Screenshots

- Home page showing 2,038 customers, 42 features, and 2 segments.

- Customer Prediction page showing the loaded model and a successful prediction.
- Customer Segments page showing a selected segment profile.
- Marketing Strategy page showing Day 3 recommendations.
- Day 4 QA table showing 6 PASS and 0 FAIL.
- Download page showing Final Dataset.csv, kmeans_model.pkl, and day3_results.pkl.

— End of Report —

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